

Audio Geolocation Pipeline

Fusing AG-CLIP Retrieval, Species Range Priors, and Scene Cues

DopplerLab Geolocation

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The Core Question

- Given a single-channel audio recording of a natural or urban soundscape . . .
- **Can we infer where on Earth it was recorded?**
- This is **passive audio geolocation**: no GPS metadata, no video, no user location tags (default mode)
- The cues are ecological and acoustic:
 - Species vocalizations → geographic range constraints
 - Habitat soundscape → retrieval against location embeddings
 - Scene context (traffic, forest) → weak auxiliary prior

What Geolocation Means in This Pipeline

- We predict a **lat/lon hypothesis** plus a **reporting granularity** (country, region, city, coordinates)
- Outputs are stored in `outputs/predictions.json`:

Field	Meaning
latitude, longitude	Fused spherical mean of hypotheses
radius_km	Uncertainty radius (heuristic)
granularity	Trust level: country / region / city / coordinates
confidence	Score at the reported granularity (not calibrated accuracy)
evidence[]	Which model branches contributed

- When `granularity_fallback` is true, prefer the **label level** over precise coordinates

Why Multi-Branch Fusion, Not One Classifier?

- No single pretrained model reliably maps arbitrary audio → GPS at city scale
- Published audio geo models on iNatSounds report **median errors of hundreds to thousands of km**
- Different branches provide **complementary constraints**:

Audio → $\underbrace{\text{retrieval}}_{\text{AG-CLIP} + \text{GeoCLIP}} + \underbrace{\text{range priors}}_{\text{SINR} / \text{eBird} / \text{GBIF}} + \underbrace{\text{scene cues}}_{\text{CLAP (low weight)}} \rightarrow \text{spherical fusion}$

- Species ID **excludes impossible regions** more often than it pins a city

Pipeline Architecture (Inference Only)

- Three parallel input branches feed into a single fusion step:
 - **AG-CLIP (22 kHz mel)** → GeoCLIP-index two-pass K-NN
 - **BirdNET + Perch** → SINR / eBird / GBIF range prior
 - **CLAP scene/text** → auxiliary scene cue signal
- All three branches converge at **spherical fusion + confidence ladder**
- Final output written to `predictions.json`
- **Scope:** frozen pretrained weights only — no training loop in this repository

Stage 1: Primary Geolocation via Retrieval

AG-CLIP (AudioGeo-CLIP)

MobileNet-V3 on mel spectrograms $\rightarrow$ 512-d geo embedding + 5,547-d species head. Trained on **iNatSounds** (CVPR 2025 benchmark).

GeoCLIP Location Encoder

Used as a **frozen GPS embedding function** for gallery indexing, not as an audio model. Hierarchical Random Fourier location encoder (NeurIPS 2023).

Retrieval Mechanism

Encode audio with AG-CLIP $\rightarrow$ cosine K-NN over $\sim 7k$ gallery points whose embeddings are GeoCLIP location codes. Two-pass: coarse global search, then masked refine using species-derived region hints.

Stage 2: Species ID and Range Constraints

Identification models

- BirdNET 2.4
- Perch 2.0 (optional)
- AG-CLIP species head (5,547 taxa)

Range prior sources

- SINR joint species-at-location model
- eBird API (optional)
- GBIF occurrence fallback

Role in fusion

- Merge detections by scientific name (max confidence)
- Build **joint range mask** over plausible lat/lon
- Emit centroid and radius as a **weighted prior signal**
- Reweight or exclude retrieval candidates outside range

Not a pin-point locator: strongest when species disagree with a wrong hemisphere hypothesis.

Stage 3: Auxiliary Scene Cues and Late Fusion

CLAP (HTSAT-fused) – auxiliary only

Zero-shot audio-text embeddings for urban/forest scene scores and geographic text prompts. Weight capped at **0.10** in audio-only fusion – must not dominate geo inference.

Spherical Fusion (`fusion.py`)

Weighted mean on the unit sphere over branch hypotheses; dynamic rules for urban scenes and species diversity.

Signal (audio-only default)	Weight
AG-CLIP two-pass retrieval	0.40
SINR range mask	0.30
CLAP text retrieval	0.10
eBird / GBIF priors	0.10 each

Published Performance of Core Models (1)

Model	Benchmark	Reported metrics		
AG-CLIP	iNatSounds test	Median error	1,082 km;	
		acc@25 km	6.4%,	@200 km
		17.2%,	@750 km	41.0%,
		@2,500 km	71.2%	
GeoCLIP	Im2GPS3k (image geo)	Acc@25 km	34.5%,	@200 km
		50.7%,	@750 km	69.7%
		(NeurIPS 2023)		
SINR	iNatSounds (ablation)	Acc@25 km	2.2%,	@200 km
		17.1%,	@2,500 km	68.6%

Metrics are from original papers – **not** re-measured on our 3-clip challenge set.

Published Performance of Core Models (2)

Model	Benchmark	Reported metrics
BirdNET	Munich urban monitoring	F1 0.46–0.84 depending on overlap / sensitivity; up to 0.84 vs. experts when tuned
Perch 2.0	BirdSet / BEANS	AUROC 0.908 , cmAP 0.431 , top-1 acc 0.665 (SOTA, no fine-tuning)
CLAP	Zero-shot audio cls.	ESC-50 93.7% (HTSAT-fused); original CLAP ZS 82.6% on ESC-50

Metrics are from original papers – **not** re-measured on our 3-clip challenge set.

Per-Clip Processing Flow

- **Preprocess** – mono mixdown; resample per model (22 kHz AG-CLIP, 48 kHz CLAP, 32 kHz Perch)
- **AG-CLIP encode** – geo vector + top species checklist indices
- **Species ID** – BirdNET CSV + optional Perch windows; European / NA heuristics for region bbox
- **Range signals** – SINR joint mask centroid; optional eBird / GBIF
- **Scene** – CLAP urban / forest / wetland softmax (auxiliary)
- **Retrieval** – AG-CLIP two-pass K-NN; CLAP text-geo prompts (low weight)
- **Fusion** – spherical mean, label selection, granularity fallback, `confidence_by_level`

What One Clip Can and Cannot Resolve

Works reasonably for

- Continent / country when species + retrieval agree
- Region-level bias (e.g. Western Europe vs. Eastern US)
- Excluding hemispheres via range masks
- Auditable evidence trail per prediction

Limited / fragile

- City-level GPS from audio alone
- Synthetic gallery not equal to real habitat prototypes
- CLAP text prompts can leak place names into city labels
- Confidence is heuristic, not calibrated
- No built-in eval harness vs. ground-truth GPS yet

Example (audio-only, 3 challenge clips): Netherlands $\times 2$ at **region**; US at **country**; mean confidence **0.585**.

Setup and Execution

- **Run from repo root:**

```
python scripts/run_pipeline.py
```

Defaults: data/ input, outputs/ results

- **Custom paths:** `--data my_clips --out my_results`
- **Required artifacts:** `ag_clip.pt`, `sinr_geo_model.pt`, FAISS gallery, taxonomy cache, CLAP HF cache
- **Optional:** `EBIRD_API_KEY`, `--use-metadata` (title/filename geocoding – not for blind eval)
- **Terminal UX:** quiet third-party logs; Preparing + Inferring progress bars; summary table at end

Evaluation Metrics (Target Framework)

- **Primary geo score** (when GT GPS available): Haversine error and threshold accuracies aligned with iNatSounds:

$$d = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_1 \cos \phi_2 \sin^2 \frac{\Delta\lambda}{2}}$$

Thresholds: 25 km (city), 200 km (region), 750 km (country), 2,500 km (continent)

- **Granularity accuracy:** fraction where reported level matches GT admin boundary
- **Species branch ablation:** fusion with vs. without SINR / BirdNET / Perch
- **Key claim to test:** does multi-branch fusion beat AG-CLIP retrieval alone at region/country scales?

Challenge Clip Results (1)

Clip	Coordinates	Prediction	Gran.	Confidence
1	52.64N, 5.08E	Netherlands	region	country 0.789 / region 0.637
2	53.31N, 4.25E	Netherlands	region	country 0.601 / region 0.540

- Both clips place within **80 km of each other** along the Dutch North Holland coast
- Clip 1 (Rotterdam anchor): AG-CLIP sim=0.134, SINR range from **17 taxa**; CLAP top prompt: suburban park birds and road noise, Netherlands
- Clip 2 (Amsterdam anchor): AG-CLIP sim=0.035, SINR range from **14 taxa**; CLAP top prompt: urban park birdsong with traffic, Amsterdam
- City-level confidence falls to 0.353 and 0.342 respectively – granularity fallback reports **region**, not city

Challenge Clip Results (2)

Clip	Coordinates	Prediction	Gran.	Confidence
3	44.47N, 73.15W	United States	country	country 0.579 / region 0.329

- Coordinates place near **Burlington, Vermont** – consistent with northeastern US species ranges
- AG-CLIP retrieval strongest of the three clips (sim=0.653, n=20 masked); SINR range from **14 taxa**
- CLAP top prompt returns *winter bird sanctuary soundscape, Netherlands* – hemisphere conflict; species signal from AG-CLIP and SINR overrode it
- Region confidence drops to 0.329; pipeline correctly **falls back to country** granularity
- radius_km widens to 1012.5 vs. 800.0 for clips 1 and 2, reflecting lower spatial agreement across branches

Roadmap and Extension Points

- **Now:** Pretrained multi-branch inference on arbitrary audio folders
- **Next:** `eval_predictions.py` + labeled manifest for measured Haversine / threshold metrics
- **Then:** Replace synthetic gallery with geo-tagged iNatSounds audio prototypes
- **Then:** Document canonical AG-CLIP checkpoint release; reduce CLAP fusion weight further
- **Later:** Spatiotemporal aggregation across neighboring recordings (paper reports up to **25.9%** @200 km with 36 km² neighborhoods)
- Each extension reuses the same fusion shell; branches swap without re-architecting the pipeline

Key References and Links

- Chasmai et al. **Audio Geolocation: A Natural Sounds Benchmark** (CVPR 2025) – arxiv 2505.18726
- Cepeda et al. **GeoCLIP** (NeurIPS 2023) – arxiv 2309.07798
- Hamilton et al. **SINR: Species Range Estimation** (NeurIPS 2024) – openreview 7CyXgo9tys
- Kahl et al. **BirdNET** (JOSS) + BirdNET-Analyzer (GitHub: birdnet-team)
- van Merrienboer and Denton, **Perch 2.0** (2025) – arxiv 2508.04665
- Wu et al. **CLAP** (ICML 2023) – arxiv 2206.04769; HF: laion/clap-htsat-fused
- Code: **nat-sound2loc** (github.com/cvl-umass/nat-sound2loc-code)